
CONFIDENTIAL MEDICAL RECORD

PATIENT INFORMATION:

- **Name:** Rajesh Kumar Menon
- **Age:** 47
- **Sex:** Male
- **Date of Birth:** 15-03-1979
- **Hospital ID:** MH-2024-89456
- **Admission Date:** 08-01-2026
- **Admission Time:** 14:30 hours

CHIEF COMPLAINTS:

- Severe headaches for 3 weeks (8/10 intensity)
- Blurred vision and occasional double vision
- Nausea and vomiting (5-6 times daily)
- Loss of balance and coordination
- Weakness in left arm and leg
- Confusion and memory lapses

HISTORY OF PRESENT ILLNESS:

Patient is a 47-year-old male who presented to the emergency department with progressive headaches starting 3 weeks ago. Initially, headaches were mild (3/10) and intermittent, but have now become constant and severe (8/10 intensity). Patient reports experiencing sudden episodes of blurred vision and diplopia (double vision) over the past 10 days. Two weeks into illness, patient developed morning nausea and vomiting, particularly worse in the mornings. This symptom has progressed to occur 5-6 times daily with projectile vomiting noted on two occasions. Five days ago, patient noticed loss of balance and coordination, making it difficult to walk. Patient reports left-sided weakness affecting both arm and leg. Wife reports patient has been increasingly confused and forgetful (forgetting recent conversations, misplacing objects). No previous history of migraines or chronic headaches. Patient denies recent head trauma, fever, or neck stiffness (argues against meningitis).

PAST MEDICAL HISTORY:

- Hypertension (diagnosed 8 years ago, on Amlodipine 5mg daily)

- Type 2 Diabetes (diagnosed 5 years ago, on Metformin 1000mg twice daily)
- Hyperlipidemia (on Atorvastatin 20mg daily)
- No known drug allergies (NKDA)

FAMILY HISTORY:

- Father: Brain tumor (astrocytoma) - died at age 62
- Mother: Diabetes Type 2
- Siblings: One brother with hypertension
- No known genetic cancer syndromes

PHYSICAL EXAMINATION:

- **General:** Alert but confused, in pain
- **Vital Signs:**
 - Blood Pressure: 158/92 mmHg (elevated)
 - Heart Rate: 92 bpm (normal)
 - Respiratory Rate: 18 breaths/min (normal)
 - Temperature: 37.1°C (normal)
 - Oxygen Saturation: 98% on room air
- **Neurological Examination:**
 - Consciousness: Alert but confused
 - Cranial Nerves: CN II-III affected (blurred vision, ptosis right eye)
 - Motor: Left arm weakness (3/5 power), Left leg weakness (3/5 power)
 - Sensation: Normal bilateral
 - Reflexes: Hyperreflexia on left side
 - Romberg Test: Positive (loss of balance)
 - Gait: Unsteady, veering to left

LABORATORY INVESTIGATIONS:

- **Complete Blood Count (CBC):**
 - Hemoglobin: 13.2 g/dL (normal: 13.5-17.5)
 - WBC: $11.8 \times 10^3/\mu\text{L}$ (elevated, normal: 4.5-11)
 - Platelets: $245 \times 10^3/\mu\text{L}$ (normal)
- **Biochemistry:**
 - Glucose (fasting): 178 mg/dL (elevated)
 - Urea: 48 mg/dL (elevated)
 - Creatinine: 1.1 mg/dL (normal)
 - Sodium: 128 mmol/L (LOW - hyponatremia due to SIADH)
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- Total Bilirubin: 0.8 mg/dL (normal)
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IMAGING STUDIES:

- **CT Brain (08-01-2026, 16:45):**
 - Large heterogeneous mass lesion in right frontal lobe measuring approximately 3.2 cm × 2.8 cm × 3.1 cm
 - Mass shows mixed density with areas of necrosis
 - Significant mass effect with compression of right lateral ventricle
 - Midline shift approximately 8 mm to the left
 - Perifocal edema (brain swelling) surrounding the mass
 - No hemorrhage noted
 - Impression: Right frontal lobe mass lesion, likely neoplastic (tumor). DDX: Glioblastoma vs High-grade Astrocytoma
- **MRI Brain with Contrast (09-01-2026, 10:30):**
 - FLAIR Sequence: Hyperintense signal in right frontal lobe
 - T1 with Gadolinium: Heterogeneous enhancement of the lesion
 - T2 Sequence: Mixed signal intensity with surrounding edema
 - DTI (Diffusion Tensor Imaging): Restricted diffusion within mass
 - Perfusion study: Elevated cerebral blood volume (CBV) and flow (CBF)
 - Corpus Callosum: No infiltration
 - Leptomeninges: No enhancement (no meningeal involvement)
 - Ventricles: Compressed right lateral ventricle
 - Final Impression: Right frontal glioblastoma (WHO Grade IV) with significant mass effect and edema. Concerning features for malignancy.

LUMBAR PUNCTURE (Cerebrospinal Fluid Analysis):

- **Performed:** 09-01-2026
- **Appearance:** Clear (no blood or cloudiness)
- **Cell Count:** WBC 8/μL (normal), RBC 0/μL
- **Glucose:** 38 mg/dL (low)
- **Protein:** 78 mg/dL (elevated, normal <45)
- **Chloride:** 112 mmol/L (normal)
- **Bacteria:** None
- **Viral PCR:** Negative

- **Cytology:** No abnormal cells seen
- **Impression:** Inflammatory profile consistent with intracranial mass

ASSESSMENT & CLINICAL IMPRESSION:

47-year-old male with:

1. RIGHT FRONTAL LOBE GLIOBLASTOMA (WHO Grade IV)

- **Evidence:**

- Imaging findings: 3.2cm heterogeneous mass with enhancement
- Clinical presentation: Localized neurological deficits
- Elevated CBV/CBF on perfusion studies
- Hyperintense on FLAIR, heterogeneous on T1/T2
- Restricted diffusion (DTI positive)

2. INTRACRANIAL HYPERTENSION

- **Evidence:**

- Headache, vomiting, confusion
- CT/MRI: 8mm midline shift, compressed ventricle
- Perifocal edema surrounding mass
- Low serum sodium (128) due to SIADH

3. MASS EFFECT MANIFESTATIONS

- **Evidence:**

- Left arm/leg weakness (motor cortex compression)
- Blurred vision (CN involvement)
- Ataxia/loss of balance (pressure on cerebellum)
- Cognitive changes (frontal lobe involvement)

DIFFERENTIAL DIAGNOSIS:

1. Glioblastoma Multiforme (Grade IV Astrocytoma) - PRIMARY

- **Confidence:** 92% (imaging + clinical presentation + CSF findings)
- **Reasoning:** Classic imaging features, rapid progression, high CBV/flow, restricted diffusion

2. High-grade Oligodendroglioma - 28%

- **Reasoning:** Can present similarly but usually slower progression, less aggressive enhancement pattern

3. Brain Metastasis (from unknown primary) - 15%

- **Reasoning:** Single lesion makes primary tumor unlikely, would expect multiple lesions

CRITICAL FINDINGS SUMMARY:

- 🚨 URGENT NEUROSURGERY CONSULTATION REQUIRED 🚨
- 🚨 URGENT ONCOLOGY CONSULTATION REQUIRED 🚨

- 🚨 High risk of cerebral herniation 🚨
- 🚨 ICP (Intracranial Pressure) monitoring needed

RECOMMENDED MANAGEMENT PLAN:

- **IMMEDIATE (First 24 hours):**
 - Neurosurgery consultation for surgical resection candidacy
 - Initiate dexamethasone 4mg IV Q6H to reduce cerebral edema
 - Osmotic therapy: Mannitol 1g/kg IV for ICP control
 - Continuous neurovital monitoring (neuro-checks Q1H)
 - ICU admission
- **SHORT TERM (Days 2-5):**
 - Neurosurgical biopsy/resection if feasible
 - Histopathology confirmation
 - Molecular testing (IDH, MGMT, TP53 status)
 - Oncology consultation for chemotherapy/radiation planning
- **LONG TERM:**
 - Post-operative radiotherapy (60 Gy in 30 fractions)
 - Temozolomide chemotherapy (concurrent and adjuvant)
 - Close imaging follow-up (MRI at 2 weeks post-op, then monthly)
 - Rehabilitation for neurological recovery

PATIENT COUNSELING:

Discussed with patient and wife:

- ✓ Diagnosis of brain tumor (glioblastoma)
- ✓ Need for urgent neurosurgery
- ✓ Risks and benefits of surgery
- ✓ Prognosis and survival statistics
- ✓ Treatment plan overview
- ✓ Support group referrals
- ✓ Advance care planning

Patient and wife understand the severity and have consented to proceed with neurosurgical evaluation.

Report Generated: 09-01-2026, 15:30 hours
By: Dr. Arun Sharma, MD (Neurology)
Institution: Apollo Hospitals, Mumbai
License No: MCI-2015-45678

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